

Arduino Based Audio VU Meter



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Presented here is an Arduino based audio volume unit (VU) meter that uses a liquid crystal display (LCD). A VU meter or standard volume indicator (SVI) is a device that displays a representation of signal level in audio equipment. In this project, the intensity of left-channel and right-channel audio signals provided as input to Arduino UNO board is displayed as bars on the 16x2 LCD. In this circuit, analogue input pins of Arduino UNO board are used for measurement of audio-signal levels.

Audio-signal inputs are provided at analogue input pins A2 and A4 of Arduino UNO board. These can be in the form of voice coming from a microphone through an amplifier or as direct output from a music player.

Audio signals at pins A2 and A4 are processed by ATmega328 microcontroller (MCU) on Arduino UNO board and, after comparing the signals, calculations are done.

Finally, corresponding values are provided by ATmega328 to a 16x2 LCD for displaying audio intensity bars.

Circuit and working

The Arduino based audio VU meter, as shown in Fig. 1, is built around Arduino UNO board, 16x2 LCD and 10-kilo-ohm potentiometer (or pot-meter).

Arduino UNO board. Arduino UNO is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. Arduino UNO board is based on ATmega328 MCU. It has 14 digital input/output (I/O) pins of which six can be used as pulse width modulated (PWM) outputs, six analogue in-

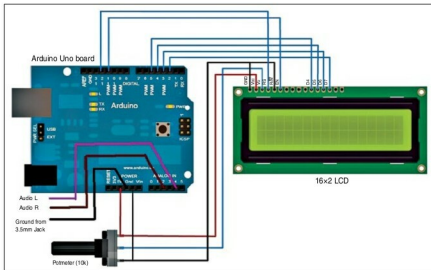


Fig. 1: Arduino based audio VU meter

puts, 16MHz ceramic resonator, USB connection, power jack, reset button and an ICSP header. It contains everything needed to support the MCU; simply connect it to a computer using a USB cable, or power it with an AC-to-DC adaptor or a battery to get started.

Pins 11 and 12 of Arduino UNO board are connected to pins EN and RS of the LCD, while pins 2 through 5 are connected to data pins D7, D6, D5 and D4 of the 16x2 LCD.

LCD 16x2. LCD JHD162A is a display module with a 4-bit interface, which is used here to display the bars corresponding to the applied input signal level. Only four data pins of the LCD module are being used in this circuit. A potmeter of 10-kilo-ohm is used to control the contrast of the LCD screen display.

Analogue audio signals (left-channel and right-channel) are provided at analogue input pins A4 and A2 of Arduino UNO board. The signals are processed by ATmega328 MCU on Arduino UNO board and, after performing some comparisons,

data is sent to the LCD screen to display the corresponding signal level bars.

Height of the bars will change as per voltage of the audio-input signal at pins A2 and A4 of Arduino UNO board as shown in Fig. 2. L is displayed on the LCD for left channel and, similarly, R is displayed for right channel. Connections of pins A2 and A4 of Arduino UNO board must be done carefully so that proper channel audio input is provided to the circuit.

To display bars on the LCD screen, custom characters are created for the 16x2 LCD, which is explained in the programming section. The potmeter is used for controlling the contrast of the display and, so, can be varied as per requirement.

Software program

The program for this circuit performs the following functions:

1. Initialising the circuit and LCD
2. Displaying the welcome message on the LCD
3. Generating custom characters
4. Performing various compari-